SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2– CASE STUDY ANALYSIS

Course Code:	CSA6501	**Course Title:**	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL2, BL3)	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	10% of CIA	**Total Marks:**	2 * 50 = 100 (2 Questions)
CO2 Statement:	*Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems (BL2, BL3)*		
Student Name:	A.Sai Rohit	**Reg. No.:**	192472144
Section / Batch:	CSE-AI	**Date:**	

Instructions to Students

- Answer **2 questions**. Each question carries **50 marks**. **Total: 100 Marks.**
- Demonstrate a **deep understanding of the concepts** relevant to the given case study.
- Identify and analyze the **key problems, requirements, challenges, and constraints** presented in the case.
- Apply appropriate **technical concepts, methods, and approaches** to propose and justify a suitable solution.
- Discuss the **real-world implications, applications, benefits, limitations, and challenges** of the proposed solution.
- Provide **thorough, evidence-based, and well-structured analysis**, supported by suitable examples, diagrams, architectures, or workflows wherever applicable.
- Present the analysis with a **clear and professional structure**, using appropriate headings and logical organization.
- Use **relevant references and proper citation practices** and maintain clear academic and technical writing.

Code of Conduct

*I _**A.Sai Rohit (192472144)**___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **A.Sai Rohit**_________________________

SECTION A – CASE STUDY ANALYSIS

3. Domain-Specific Legal Assistant

A legal organization wants to adapt a general-purpose LLM to understand legal terminology and generate responses in a consistent legal format.

Analyze whether full fine-tuning, transfer learning, few-shot prompting, or PEFT/LoRA would be most appropriate. Discuss the advantages and limitations of your selected approach.

Introduction

Large Language Models (LLMs) such as GPT, Llama, and Mistral have revolutionized Natural Language Processing (NLP) by generating human-like text and answering complex questions. However, these models are trained on general-purpose datasets and may lack the specialized knowledge, terminology, and formatting required in legal practice.

A legal organization intends to adapt a general-purpose LLM so that it understands legal terminology, interprets legal documents accurately, and generates responses in a standardized legal format. Choosing the appropriate adaptation technique is essential because legal applications require high accuracy, consistency, privacy, and cost efficiency.

Among the available approaches—Full Fine-Tuning, Transfer Learning, Few-Shot Prompting, and PEFT (Parameter-Efficient Fine-Tuning) using LoRA (Low-Rank Adaptation)—PEFT/LoRA is the most appropriate solution because it offers high performance with significantly lower computational cost while preserving the knowledge of the original model.

1. Problem Statement

The legal organization wants to develop an AI-powered legal assistant capable of:

- Understanding legal terminology and statutes.
- Answering legal queries accurately.
- Drafting legal documents in a consistent format.
- Summarizing legal judgments.
- Maintaining confidentiality of legal information.
- Reducing training and deployment costs.
- Updating knowledge efficiently as laws evolve.

Since legal decisions can have serious consequences, the model must provide reliable, consistent, and well-structured responses.

2. Requirements Analysis

Functional Requirements

- Interpret legal terminology correctly.
- Answer legal questions.
- Generate contracts and legal notices.
- Summarize case laws.
- Maintain standardized legal formatting.
- Retrieve relevant legal information.

Non-Functional Requirements

- High accuracy
- Data privacy and confidentiality
- Fast response time
- Low computational cost
- Scalability
- Easy maintenance

- Regulatory compliance

3. Analysis of Available Approaches

A. Full Fine-Tuning

Description

Full fine-tuning retrains all parameters of the pretrained LLM using legal datasets.

Advantages

- Highest level of customization.
- Excellent understanding of legal terminology.
- Produces highly accurate responses.
- Learns legal writing style completely.

Limitations

- Very expensive.
- Requires multiple high-end GPUs.
- Long training time.
- High energy consumption.
- Difficult to update frequently.
- Risk of overfitting.

Suitability

Suitable only for organizations with abundant computational resources and very large legal datasets.

B. Transfer Learning

Description

Transfer learning starts with a pretrained model and retrains it on legal datasets.

Advantages

- Better than training from scratch.
- Requires less data.
- Good adaptation to legal language.
- Better accuracy than prompt engineering.

Limitations

- Still computationally expensive.
- Updates many parameters.
- Requires specialized hardware.

Suitability

Appropriate for medium-sized organizations but still expensive compared to PEFT.

C. Few-Shot Prompting
Description

Few-shot prompting teaches the model by providing examples within the prompt without updating model parameters.

Advantages

- No training required.
- Fast implementation.
- Very low cost.
- Easy experimentation.

Limitations

- Inconsistent responses.
- Limited understanding of legal terminology.
- Prompt length restrictions.
- Cannot permanently learn legal knowledge.

Suitability

Useful for prototyping but not suitable for production-level legal assistants.

D. PEFT (Parameter-Efficient Fine-Tuning) with LoRA (Selected Approach)
Description

PEFT updates only a small number of additional parameters while keeping the original pretrained model frozen. LoRA introduces low-rank matrices into selected layers of the transformer instead of retraining all billions of parameters.

This approach enables efficient learning of legal terminology and document formatting while preserving the general language understanding of the original model.

4. Comparison of All Approaches

Criteria	Full Fine-Tuning	Transfer Learning	Few-Shot Prompting	PEFT/LoRA
Training Cost	Very High	High	Very Low	Low
GPU Requirement	Very High	High	None	Low
Training Time	Very Long	Long	None	Short
Memory Usage	Very High	High	Low	Low
Domain Adaptation	Excellent	Good	Moderate	Excellent
Maintenance	Difficult	Moderate	Easy	Easy
Scalability	Low	Medium	High	High
Best for Legal Assistant	No	Moderate	No	**Yes**

5. Justification for Selecting PEFT/LoRA

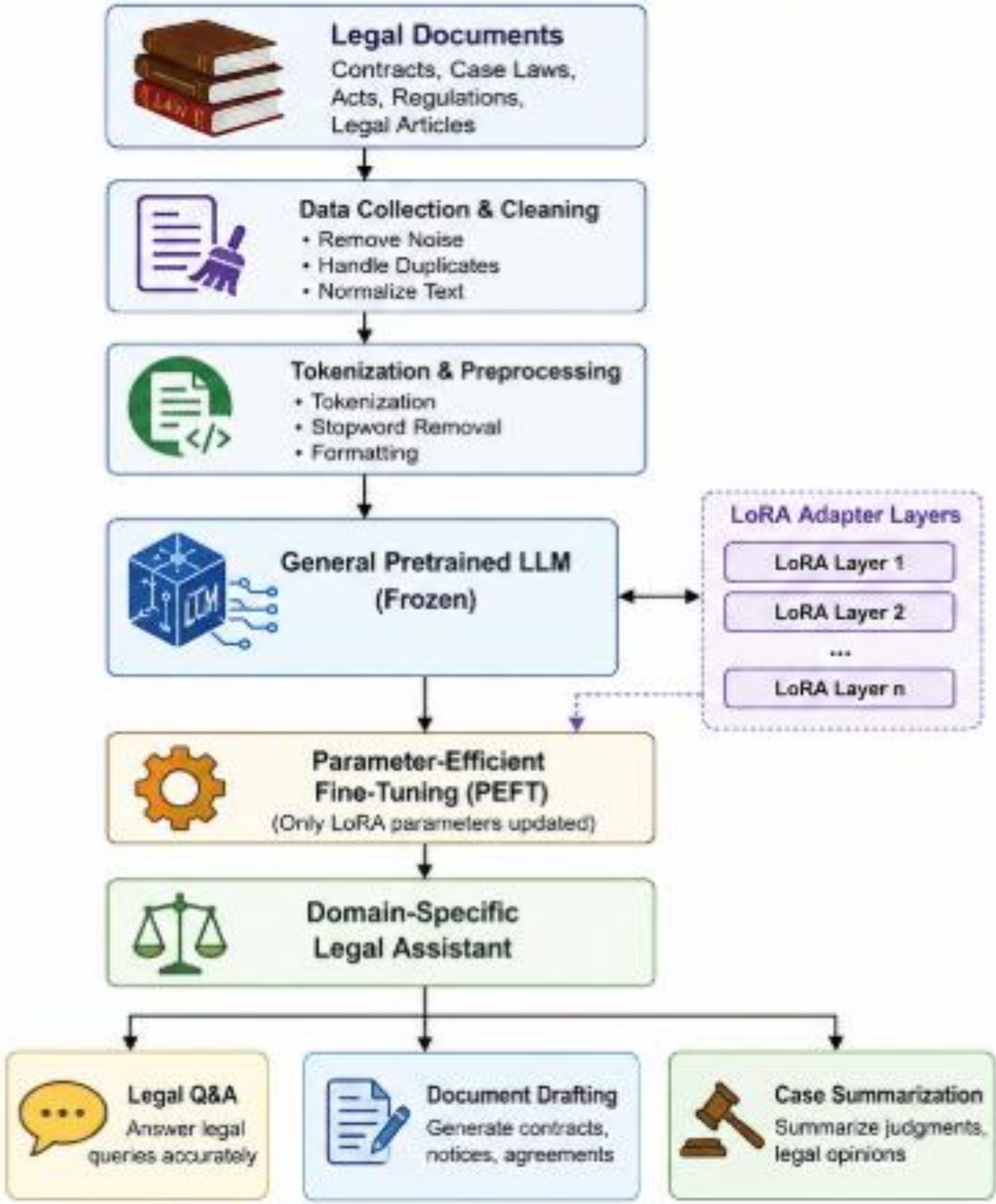
PEFT with LoRA is the most appropriate solution because it balances performance, cost, and efficiency.

Reasons

- Learns legal terminology effectively.
- Produces consistent legal document formatting.
- Preserves the knowledge of the pretrained model.
- Requires significantly less computational power.
- Supports rapid updates as laws change.
- Enables multiple legal specializations using separate LoRA adapters.
- Reduces deployment costs.

Therefore, PEFT/LoRA satisfies both the technical and business requirements of the legal organization.

6. Proposed System Architecture



8. Advantages of PEFT/LoRA

High Accuracy

Learns legal terminology and legal writing style effectively.

Low Computational Cost

Only a small percentage of parameters are trained.

Faster Training

Training can be completed within hours instead of several days.

Lower Memory Usage

Suitable for organizations with limited hardware resources.

Easy Updates

Different adapters can be developed for:

- Criminal Law
- Civil Law
- Corporate Law
- Tax Law
- Intellectual Property Law

without modifying the original model.

Better Scalability

Organizations can deploy multiple legal assistants using the same base model.

9. Limitations of PEFT/LoRA

- Performance depends on the quality of the pretrained model.
- Requires high-quality legal datasets.
- May not outperform full fine-tuning in extremely specialized legal tasks.
- Cannot replace professional legal experts.
- Periodic retraining is needed when laws change.

10. Challenges and Constraints

Technical Challenges

- Large legal datasets
- Complex legal terminology
- Maintaining response consistency
- Avoiding hallucinations

Legal Challenges

- Privacy of confidential legal documents
- Compliance with legal regulations
- Ethical use of AI
- Accountability for incorrect legal advice

Operational Challenges

- Continuous model updates
- Human verification
- Secure deployment
- Cost management

11. Real-World Implications and Applications

A domain-specific legal assistant can significantly improve legal services.

Applications

- Legal research
- Contract drafting
- Case law summarization
- Compliance monitoring
- Client support
- Legal document classification
- Court judgment analysis
- Policy interpretation

Benefits

- Saves lawyers' time.
- Reduces repetitive work.
- Improves document consistency.
- Provides faster legal assistance.
- Reduces operational costs.
- Increases productivity.
- Supports legal education.

Limitations in Real World

- AI-generated responses may contain factual errors.
- Human legal experts must verify important legal advice.
- Sensitive legal data requires strong cybersecurity measures.
- Bias in training data can affect fairness.

12. Future Enhancements

Future improvements may include:

- Integration with Retrieval-Augmented Generation (RAG) for access to the latest legal databases.
- Explainable AI to justify legal reasoning.
- Multilingual legal document support.
- Continuous learning from newly enacted laws and court judgments.
- Integration with digital court systems.

13. Conclusion

General-purpose Large Language Models require adaptation before they can effectively operate in specialized domains such as law. Among the available techniques—Full Fine-Tuning, Transfer Learning, Few-Shot Prompting, and PEFT/LoRA—PEFT with LoRA is the most suitable approach for developing a domain-specific legal assistant.

It offers an excellent balance between accuracy, efficiency, scalability, and cost. By training only a small number of additional parameters while preserving the original model, PEFT enables organizations to develop high-quality legal AI systems without requiring expensive computational resources. It supports legal research, document drafting, case summarization, and client assistance while allowing easy updates as legal regulations evolve. Although human oversight remains essential due to the sensitive nature of legal decisions, PEFT/LoRA provides a practical and effective solution for deploying reliable AI-powered legal assistants.

18. Limited Computational Resources

A startup wants to customize a large LLM for customer support but has limited GPU resources and budget.

Analyze why full fine-tuning may be unsuitable. Explain how LoRA and PEFT can address the problem and recommend an appropriate adaptation strategy.

Introduction

Large Language Models (LLMs) such as GPT, Llama, and Mistral are widely used for customer support because they can understand customer queries and generate natural, human-like responses. However, startups often face significant constraints, including limited GPU resources, restricted budgets, and small technical teams. As a result, choosing the right model adaptation technique is essential.

Among the available methods, Full Fine-Tuning is not suitable due to its high computational cost. Instead, Parameter-Efficient Fine-Tuning (PEFT) using LoRA (Low-Rank Adaptation) is the most appropriate solution because it enables efficient customization while requiring minimal computational resources.

1. Problem Statement

A startup wants to customize a large language model for customer support. The AI assistant should:

- Answer customer queries accurately.
- Understand company-specific products and policies.
- Maintain a consistent response style.
- Reduce training and deployment costs.
- Operate with limited GPU resources.
- Be easy to update as business information changes.

The startup must select an adaptation strategy that balances performance, cost, and scalability.

2. Requirements Analysis

Functional Requirements

- Answer customer questions.
- Understand product information.
- Handle FAQs.
- Generate polite and professional responses.
- Support conversation history.

Non-Functional Requirements

- Low computational cost.
- Fast training.
- Low memory usage.
- High accuracy.
- Scalability.
- Easy maintenance.
- Quick deployment.

3. Why Full Fine-Tuning is Unsuitable

Description

Full fine-tuning updates all parameters of the pretrained LLM using customer support data.

Drawbacks

1. High GPU Requirement

Large LLMs contain billions of parameters and require multiple high-end GPUs.

2. Very Expensive

Training costs are beyond the budget of most startups.

3. Long Training Time

Training may take several days or even weeks.

4. High Memory Consumption

Large GPU memory is required to store gradients and optimizer states.

5. Difficult Maintenance

Every update requires retraining the complete model.

6. Risk of Overfitting

Limited customer support data may cause the model to memorize instead of generalize.

Conclusion

Therefore, Full Fine-Tuning is not suitable for startups with limited computational resources.

4. Understanding PEFT and LoRA

Parameter-Efficient Fine-Tuning (PEFT)

PEFT updates only a small subset of parameters while keeping the pretrained model frozen.

Instead of modifying billions of parameters, PEFT trains only a few million parameters.

Low-Rank Adaptation (LoRA)

LoRA is one of the most popular PEFT techniques.

It inserts small trainable matrices into selected transformer layers while leaving the original model unchanged.

Thus, only the adapter parameters are learned during training.

5. How LoRA and PEFT Solve the Problem

1. Lower Computational Cost

Only a small percentage of parameters are trained, reducing computational requirements.

2. Reduced GPU Memory

LoRA requires significantly less GPU memory compared to full fine-tuning.

3. Faster Training

Training completes much faster because only adapter parameters are updated.

4. Lower Storage Requirements

Only lightweight LoRA adapter files are stored instead of multiple complete models.

5. Easy Model Updates

Business information can be updated by retraining only the adapters.

6. Multiple Business Versions

Different LoRA adapters can be created for:

- Sales Support
- Technical Support
- Billing Support
- Product Recommendations

without changing the base model.

6. Comparison of Adaptation Approaches

Criteria	Full Fine-Tuning	Few-Shot Prompting	PEFT/LoRA
GPU Requirement	Very High	None	Low
Training Cost	Very High	Very Low	Low
Memory Usage	Very High	Low	Low
Training Time	Very Long	None	Short
Performance	Excellent	Moderate	High
Maintenance	Difficult	Easy	Easy
Startup Friendly	No	Partially	**Yes**

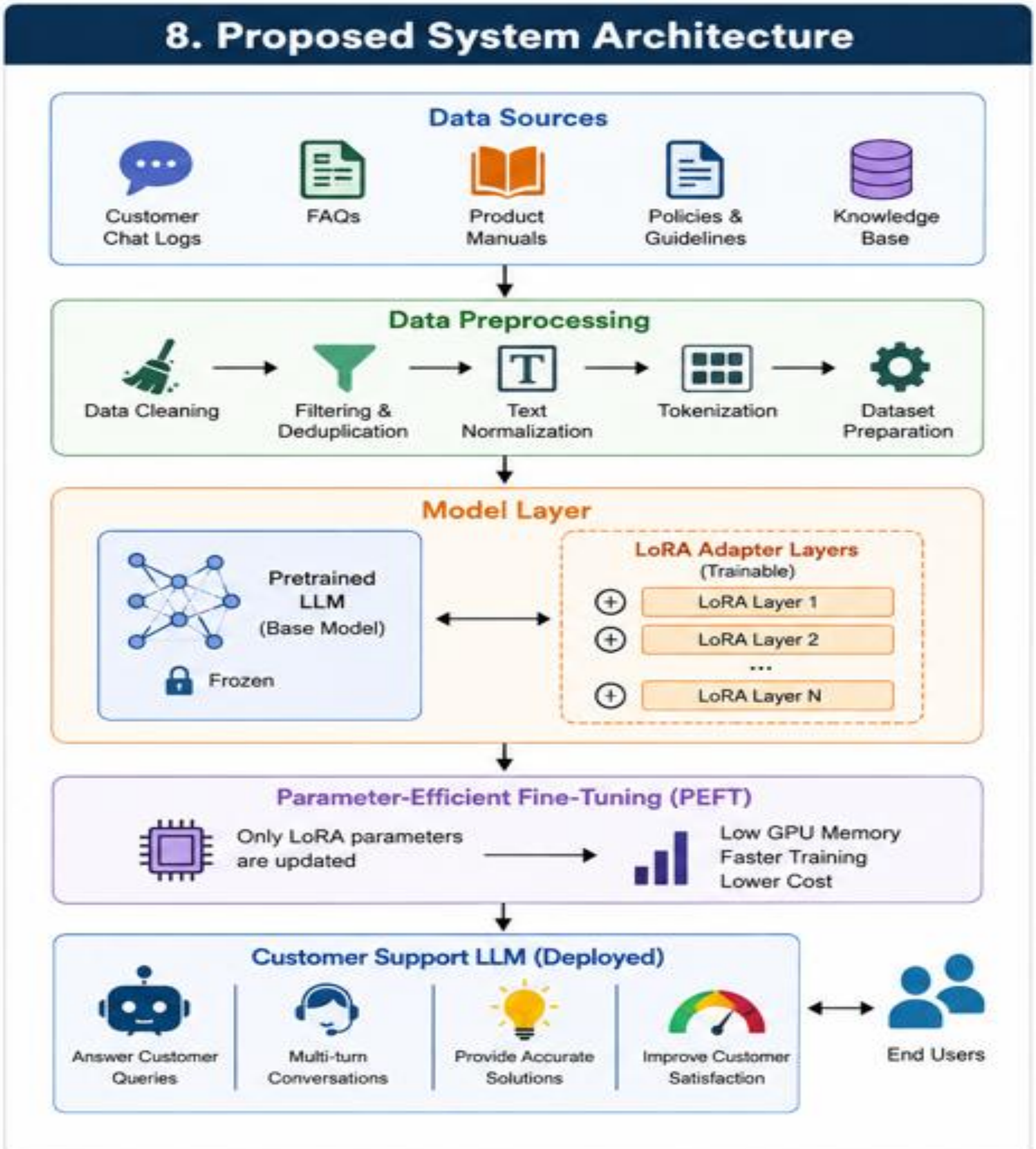
7. Recommended Adaptation Strategy

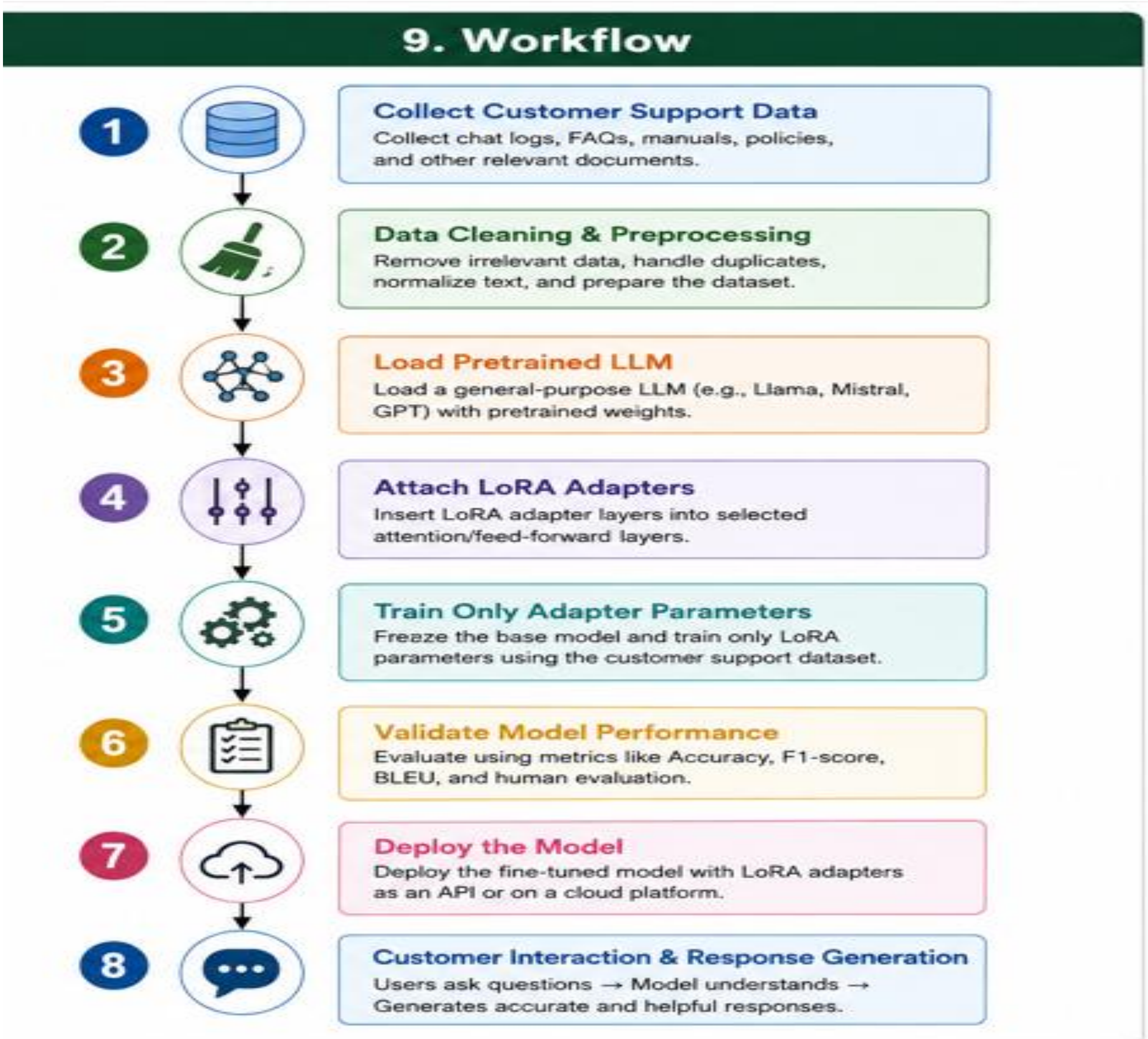
The startup should adopt PEFT with LoRA.

Justification

- Low GPU requirement.
- Low training cost.
- Faster customization.
- High-quality customer support responses.
- Easy deployment.
- Supports continuous updates.
- Suitable for startups with limited infrastructure.

This approach provides the best balance between efficiency, accuracy, and affordability.


8. Proposed System Architecture
Data Sources
Customer Chat Logs
FAQs
Product Manuals
Policies & Guidelines
Knowledge Base
Data Preprocessing
Data Cleaning
Filtering & Deduplication
Text Normalization
Tokenization
Dataset Preparation
Model Layer
Pretrained LLM (Base Model)
Frozen
LoRA Adapter Layers
(Trainable)
LoRA Layer 1
LoRA Layer 2
...
LoRA Layer N
Parameter-Efficient Fine-Tuning (PEFT)
Only LoRA parameters are updated
Low GPU Memory
Faster Training
Lower Cost
Customer Support LLM (Deployed)
Answer Customer Queries
Multi-turn Conversations
Provide Accurate Solutions
Improve Customer Satisfaction
End Users



10. Advantages of PEFT/LoRA

- Low GPU requirements.
- Faster model training.
- Lower cloud computing costs.
- High-quality customer responses.
- Easy maintenance and updates.
- Supports multiple adapters for different tasks.
- Preserves the original model's knowledge.
- Suitable for startups and small businesses.

11. Limitations

- Performance depends on the quality of the base model.
- Requires well-prepared customer support datasets.
- May not reach the absolute peak performance of full fine-tuning.
- Periodic retraining is required when products or policies change.

12. Real-World Implications

Applications

- Customer support chatbots.
- Technical support assistants.
- E-commerce help desks.
- Banking customer service.
- Healthcare appointment assistants.
- Telecom support systems.
- Travel booking assistants.

Benefits

- 24/7 customer support.
- Reduced operational costs.
- Faster response times.
- Improved customer satisfaction.
- Increased business scalability.

Challenges

- Ensuring data privacy.
- Preventing incorrect or misleading responses.
- Handling multilingual customer queries.
- Keeping knowledge up to date.

13. Future Enhancements

- Integrate Retrieval-Augmented Generation (RAG) to provide up-to-date answers from company documents.
- Support multilingual customer interactions.
- Add sentiment analysis for personalized responses.
- Include human-agent handoff for complex issues.
- Continuously improve adapters with customer feedback.

14. Conclusion

For a startup with limited GPU resources and budget, Full Fine-Tuning is impractical because it requires significant computational power, large memory capacity, long training times, and high operational costs. In contrast, PEFT with LoRA offers an efficient and cost-effective solution by updating only a small number of parameters while keeping the pretrained model unchanged. This approach provides high-quality customer support, reduces infrastructure costs, enables rapid updates, and scales easily as the business grows. Therefore, PEFT with LoRA is the recommended adaptation strategy for building an effective customer support LLM under limited computational resources.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Depth	30%	Deep, accurate understanding of pre-training/fine-tuning/RLHF concepts	Good understanding with minor gaps	Superficial understanding of concepts	Misunderstands core concepts
Real-World Implications	25%	Draws well-reasoned real-world implications and comparisons	Reasonable implications drawn	Weak or generic implications	No meaningful analysis presented
Analytical Rigor	20%	Analysis is thorough, evidence-based, and well-structured	Mostly thorough analysis	Partial analysis with gaps in evidence	Superficial, unstructured analysis
Report Structure	15%	Report/slides professionally structured, easy to follow	Clear structure with minor issues	Structure unclear in places	Poorly structured submission
Citation & Academic Writing	10%	Properly cites sources; clear academic writing	Mostly cited, minor lapses	Inconsistent citation practice	No citation or referencing

Marks Evaluation:

Question No.	Conceptual Depth(30%)	Real-World Implications (25%)	Analytical Rigor (20%)	Report Structure (15%)	Citation & Academic Writing(10%)	Total Marks (100)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________